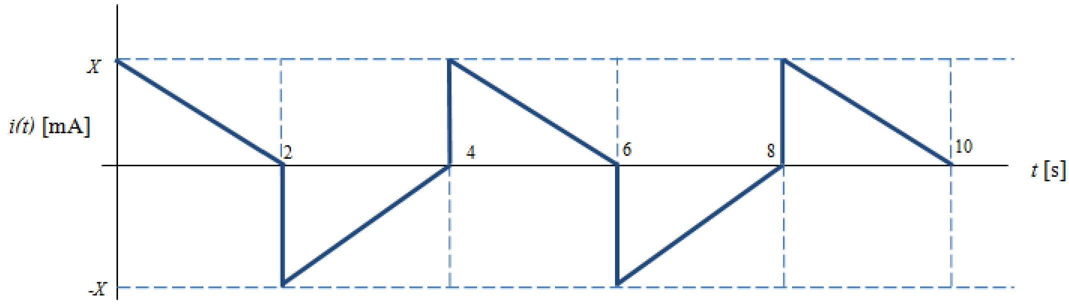
Review Test Submission: Assignment Week 1

User	Mr W (Wouter) W Nel
Course	EBN 122 S2 2017
Test	Assignment Week 1
Started	7/21/17 3:56 PM
Submitted	7/21/17 4:20 PM
Due Date	7/24/17 11:59 PM
Status	Completed
Attempt Score	15.5 out of 16 points
Time Elapsed	24 minutes
Instructions	Read over the document *ClickUP Assignments Instructions*. Save the answer to each question as you complete it. At the end of the assignment click Save and Submit. Give your final correct answer 3 significant digits. Make sure you use the FULL STOP for your decimal place marker. You must indicate the units, unless otherwise specified: • There must be NO space between the answer • There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V* • Write out 'ohm' to indicate Ω • Use '**u**' to indicate micro. • Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G)** and tera(**T**); do not use centi, deci and deca
Results Displayed	All Answers, Submitted Answers, Feedback, Incorrectly Answered Questions

Question 1

1.5 out of 2 points

The graph shows the current through an element over time. Determine the charge that passes through the element between 2 and 8 seconds for a current of 43 mA.

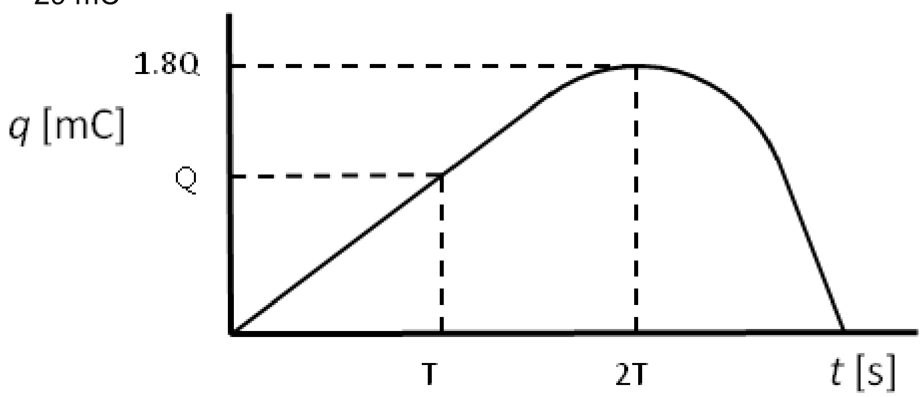


Selected Answer: -43mA

Question 2

2 out of 2 points

The charge flowing in a conductor is shown below. Determine the current at t = 2T s
(The curve reaches its maximum value at t = 2T s.)
T = 0.5 s
Q = 29 mC



Selected Answer: 0A

Question 3

2 out of 2 points

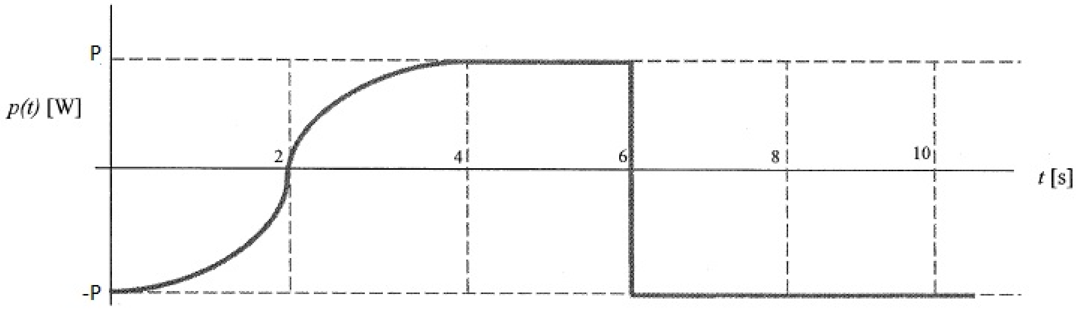
To move -65 nC in 9.8 micro seconds through an element requires 650 mJ. Determine the current through the element, assuming a constant current.

Selected Answer: -6.63mA

Question 4

2 out of 2 points

The graph below shows the power of an circuit over time. Determine the energy used by the circuit between 4 and 9 seconds. The value of P is 9.5 W.

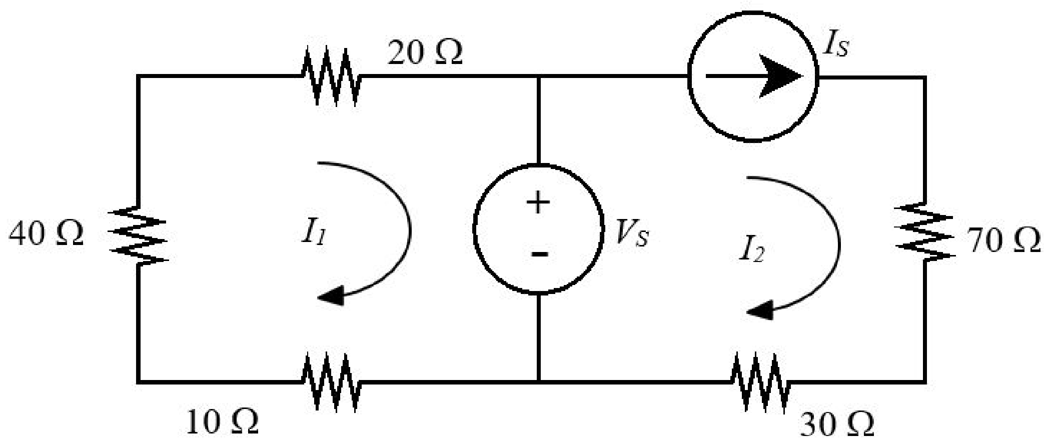


Selected Answer: -9.5J

Question 5

2 out of 2 points

How many nodes (n) **[n]**, independent loops (l) **[l]**, and branches (b) **[b]** are there in the following circuit?



Specified Answer for: n 6

Specified Answer for: l 2

Specified Answer for: b 7

Question 6

2 out of 2 points

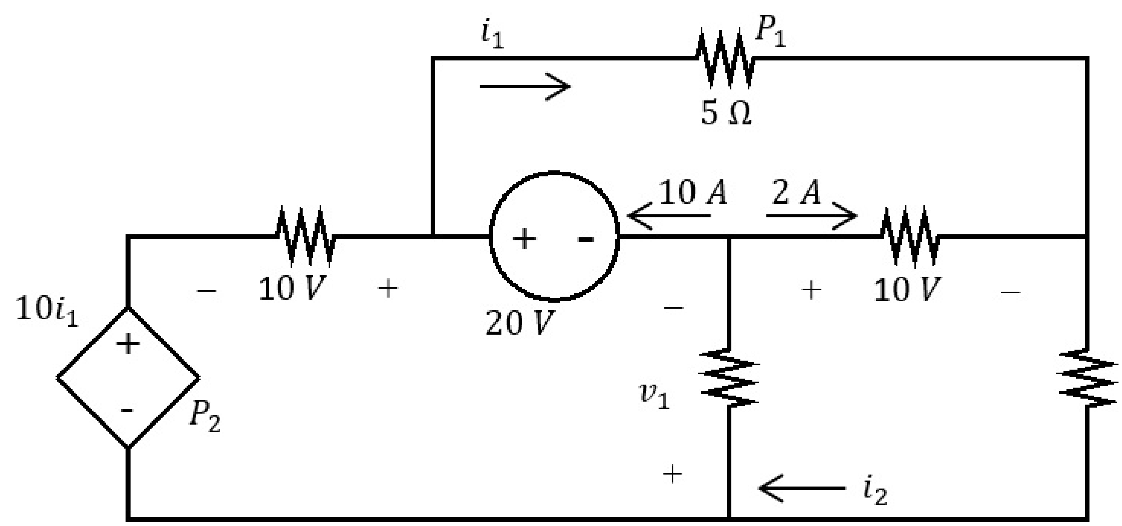
An electric heater is specified as 255 V; 2.5 kW.What is the current flowing through the heater?

Selected Answer: 9.8A

Question 7

2 out of 2 points

Determine P_1 if i_1 = 7.3 A.



Selected Answer: 266W

Question 8

0 out of 0 points

The voltage across an element is $v(t) = \frac{di}{dt}$ V and the current through the element is $i(t) = 2e^{-4t}$ mA. Determine the power $p(t)$ (in μW) (associated with the element as a function of time. The current flows from the negative to the positive potential through the element. (In other words, the current is entering the negative terminal and exiting the positive terminal.)

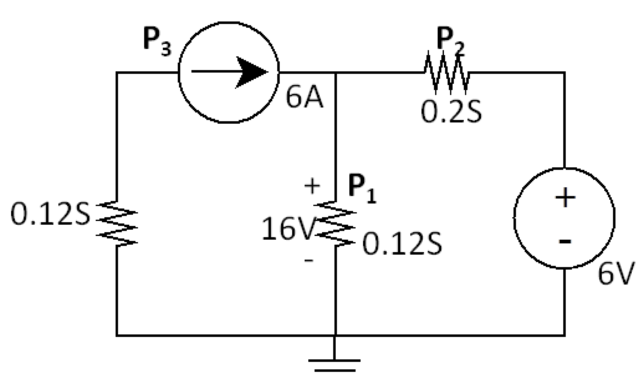
The answer is in the form Ae^{Bt} with A=**[A]** and B=**[B]**

Specified Answer for: A -16

Specified Answer for: B -8

Question 9

0 out of 0 points



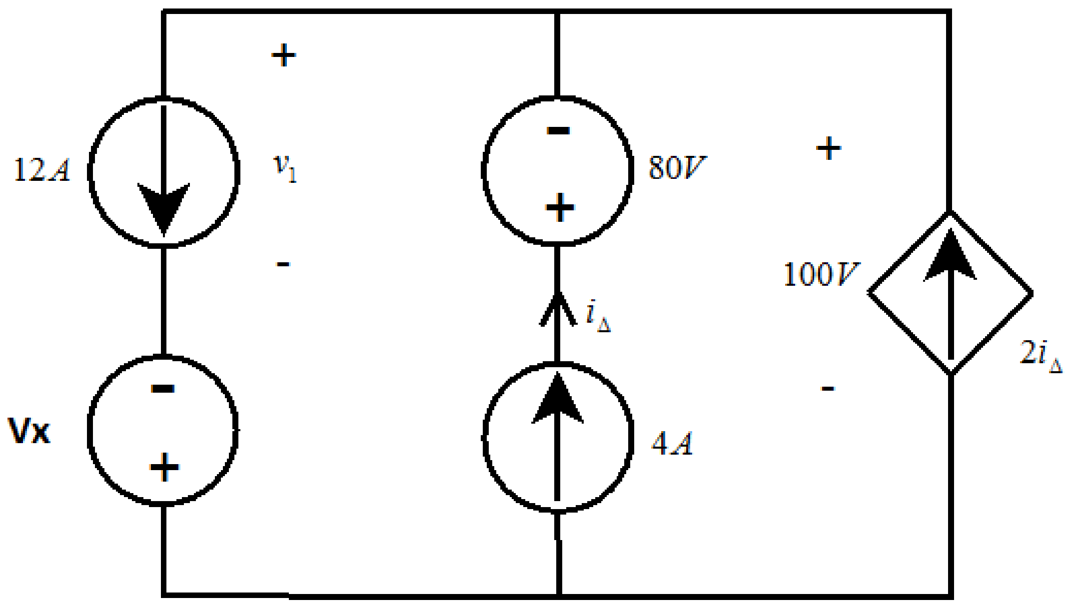
Calculate the Power P_1.

Selected Answer: 30.7W

Question 10

2 out of 2 points

Calculate the power associated with the voltage source on the left most side of the circuit if V_x = 66.



Selected Answer: -792W

Friday, July 21, 2017 4:21:07 PM SAST

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